Department of Electrical Engineering
香港城市大學
City University of Hong Kong

CityU Architecture Lab for Arithmetic and Security (CALAS) Seminar Series

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Recent Advances and Challenges in Cryptography

Prof. Huaxiong Wang, Professor
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Abstract:

Recent advancements in cryptography are driving innovation across multiple domains, yet they also introduce significant challenges. This talk will explore three key areas at the forefront of cryptographic research: privacy-preserving technologies, post-quantum cryptography, and privacy in artificial intelligence (AI). In the domain of privacy-preserving technologies, we will delve into fully homomorphic encryption, secure multiparty computation, and verifiable computation—each offering promising avenues for enabling secure data processing while preserving privacy. Post-quantum cryptography represents a vital response to the looming threat posed by quantum computing. We will discuss the ongoing NIST competition aimed at standardizing quantum-resistant algorithms, and the challenges of migrating existing systems to post-quantum standards. Lastly, privacy in AI is an emerging field of concern. We will highlight how fully homomorphic encryption can enable privacy-preserving machine learning, with a particular focus on private neural networks, showing how cryptographic tools can ensure confidentiality in AI-driven environments.

Biography:

Huaxiong Wang holds a PhD in Mathematics from the University of Haifa, Israel (1996), and a PhD in Computer Science from the University of Wollongong, Australia (2001). He is a President's Chair Professor in Cryptology at Nanyang Technological University (NTU), Singapore, where he served as the Head of the Division of Mathematical Sciences from 2013 to 2015. Currently, he also serves as Co-Director of the National Digital Trust Centre and Director of the Strategic Centre for Research in Privacy-Preserving Technologies & Systems at NTU. Before joining NTU in 2006, Prof. Wang held academic positions at Macquarie University and the University of Wollongong in Australia. With over 30 years of research experience in cryptography and information security, he has authored or co-authored one book, edited nine volumes, and published more than 300 papers in leading international journals and conferences. His work spans a wide range of topics in cryptography and information security.
Prof. Wang has supervised more than 30 PhD students, served on the editorial boards of several international journals, and participated in over 100 international conference program committees, frequently as a chair. In recognition of his contributions, he received the inaugural Best Research Contribution Award from the Computer Science Association of Australasia in 2004. He was an invited speaker at ASIACRYPT 2017 and served as Program Co-Chair of ASIACRYPT in 2020 and 2021. He was elected as Fellow of International Association for Cryptologic Research (IACR) in 2026.

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